

**ADMINISTRATIVE INFORMATION****FMI No: 25351****FMI Title: LightSpeed Pro & RT (100kW 80kW & RT) Program M4 FMI**

FMI TYPE	CHARGE CODES	IMPLEMENTATION DATES
☐ Mandatory Safety	GEMS - AM:	Issue Date:
☒ Mandatory	GEMS - E:	
☐ Mandatory on Request	GEMS - A:	Close Date:

☒ Staged FMI	Customer List provided by FMI Administration. If you identify additional units within effectivity range, but they are not shown on the customer list, contact FMI Administration in your Pole.
☒	Parts and any special tools are furnished automatically. To address shortages and ordering exceptions, contact FMI Administration in your Pole.
☐	Parts Ordering: GEMS - Americas, Contact your Region logistics coordinator GEMS - Europe, Order FMI kits from GPO/EPL Buc GEMS - Asia, Contact your region logistics coordinator
☐	No Parts are required to complete this FMI.
☐ Batch with	_____
	To minimize costs, combine implementation of this FMI with the FMI's listed for all customers where the effectivity is the same (Please see your customer lists).
☐ Non-Staged	No Customer List is provided. Each service location must identify affected units, and place part orders as indicated in the Special Instructions.

Technical Support		
GEMS - Americas	GEMS - Europe	GEMS - Asia
GE Medical Systems Insite Center (414) 524-5300 800-321-7973 Milwaukee, WI 53201, USA	GEMS - E / ESC BP34 F 78533 BUC CEDEX, FRANCE Tel: (33 1) 39 20 00 07 or ESC Fax: (33 1) 30 70 99 70	Asia Support Center Phone: 81-426-56-0033 Fax: 81-426-48-2905

Special Instructions:**Unit Tracked:****FMI Instructions Part No.: 5131410-100****Do NOT leave unsecured on site**

FMI Completion Certification Report

Mail To

GEMS - Americas	GEMS - Europe	
G.E. Medical Systems Attn: FMI Admin. W-523 P.O. Box 414 Milwaukee, WI 53201	GEMS - Europe FMI Admin. 322 BP34 F78533 Buc Cedex, France	Send to your Country FMI or Service Administrator

FMI No.: 25351

FMI Title: LightSpeed Pro & RT (100, 80kW & RT)Program M4 FMI

Customer Full Name:

Number and Street Address:

City and Country:

Country Code

Site Number

System I.D.

Unit Tracked:

Model Number	Serial Number	Compl. Date DD/MM/YY	FMI Comp. Code	Service Hours	
				Labor	Travel

FMI Completion Code

1. Modification Complete or previously modified.
2. FMI not required per FMI instruction.
3. Not modified. Unit in storage. FMI kit returned to stock.
4. Modification refused or equipment altered. Customer signature and title required.
5. Unit location unknown or unit not in area. Product Locator notified.

Customer Refusal or Equipment Altered - Modification Pending.

Installer

Comments

Effectivity

The following Milwaukee 5.X products, produced and shipped prior to March 23, 2005:

- HP100 & HP80 Hercules base (MN: 2357739, 2357739-4, 2357739-6, 2357739-7),
- HP80 Varian (MN: 2357739-3)
- RT (MN: 2374681-2)

Purpose

The purpose of this FMI is to upgrade all 5.X systems with M4 software and to perform the following system quality checks:

- Table Cradle Latch Inspection & Correction
- Table Drive Connections Inspection
- HP Table Grounding Inspection
- kV Circuit Board Inspection
- NGPDU Inspection

Common Software Platform Q1 2005 Release

04MW44.9 Overview

This software release has addressed numerous PSR, PQR and other problem reports. Please see below for new features in the release and problems resolved by the release.

New Features LightSpeed Pro16 systems for M4 release

Increased DAS sampling for 0.5 and 0.4 second scans (only available on GDAS based systems)

The DAS sampling rate has been increased for 0.5 and 0.4 second rotation times. 0.5 second sample rate is 1968 hz giving 968 views per rotation. 0.4 second sample rate is 2000 hz giving 800 views per rotation. The increased sample rate will provide improved image quality at these rotation times.

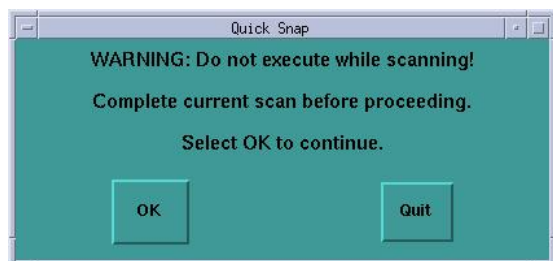
Rotor Management

Management of the tube rotor has been updated. The rotor will be allowed to be brought up once every 90 seconds. This time is shortened from every 3 minutes in the previous release. This change reduces wear on the tube rotor and extends tube life.

New Features Common Platform

Quick Snap

Quick Snap gathers critical systems files to help CT Engineering trouble shoot system problems. Customers should be directed to initiate Quick Snap when the system is in the problem state. Quick Snap should not be initiated when X-ray is on. The data from the Quick Snap is stored in /usr/g/service/log/ with an .inst extension. Quick Snap is initiated from the Tool chest in the upper right of the Image Works desktop.



Issues addressed

Reconstruction

Recon control crashes when 0 views in Cardiac Helical scan
Recon data path now reports correct FRU level

System

Tftp files fill up system disk
Power up of Linux based system shuts down a LightSpeed 1.X system
GDAS converter cards stuck in External gain calibration. Scan hardware hung. Caused by pause and resume scan while in DAS gain cal.
ScanRx process crashed
Daily Prep crashed on corrupt cal vector

Tube Warmup fails if apostrophe in Hospital name

Scanning

System aborts scan due to

- DAS calibration on GDAS systems

- Rpeak detection timeout

- Cradle not ready

- Offset collection delay due to delay in moving table to scan

- Axial not ready

Breathing lights not aligned with X-ray

Insufficient ISD to play Auto Voice

Gantry Display shows cradle latched but it is not

Table /cradle will not move in while laser lights on

Prescription

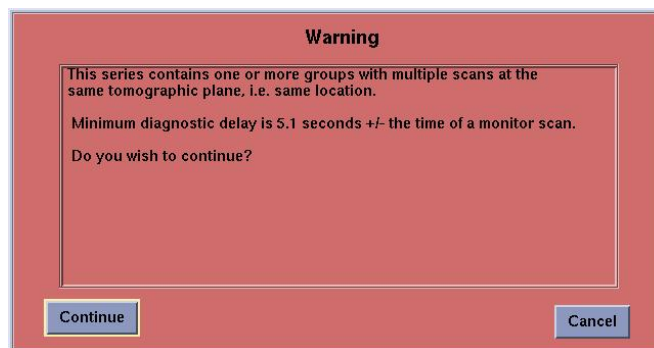
Brighter yellow color for Selected buttons to make more apparent

Additional slices added when scans are prescribed with gantry tilt

Show Localizer lines are in wrong location if user skips series quickly

SmartPrep Diagnostic delay not as expected.

The diagnostic delay is adjusted at Confirm to account for table movement required between the Monitor and Scan phase locations. If the Diagnostic Delay prescribed is less than what the system can do, the expected delay will be posted at Confirm.



SmartPrep fails to start

Start Locations have odd numbers. Adjustments to Start Locations from Graphic Rx will be rounded off in .25mm increments.

Dose displayed is higher than expected if do Next Series very quickly

Auto mA turned off if Next Series before scout available

Cardiac pitch changed to 0.24:1 for 0.4 second SnapShot Burst scans for better image quality

Display/Filming/Network/Archive

Allow 5 digit port numbers for DICOM camera set up

Clean up dcserver cores

Film jobs paused in queue even though images have filmed

Images can't be retrieved from PACS

Problem with transfer syntax for color saved images

Color Saved images filmed to Post Script printer from Exam Rx desktop are blank

Storage Commitment query slows down CT system

Dcserver crashes with Storage Commitment requests

arslog fills up the system

Images are filmed cut off

Too much annotation on filmed Dentascan images

Service Tools

Add reset Image Database capability – resetImageDB_Linux (all images will be removed from the scanner)

Remote List Select images shows date as 00/00/00 when system configured in French language

Script to convert to old CT user interface colors removed. Business directive is all Linux based system use new CUI colors.

Performix® Pro 100 CT Tube Overview

(GDAS 80, 100 and MDAS 100 configured systems)

Power

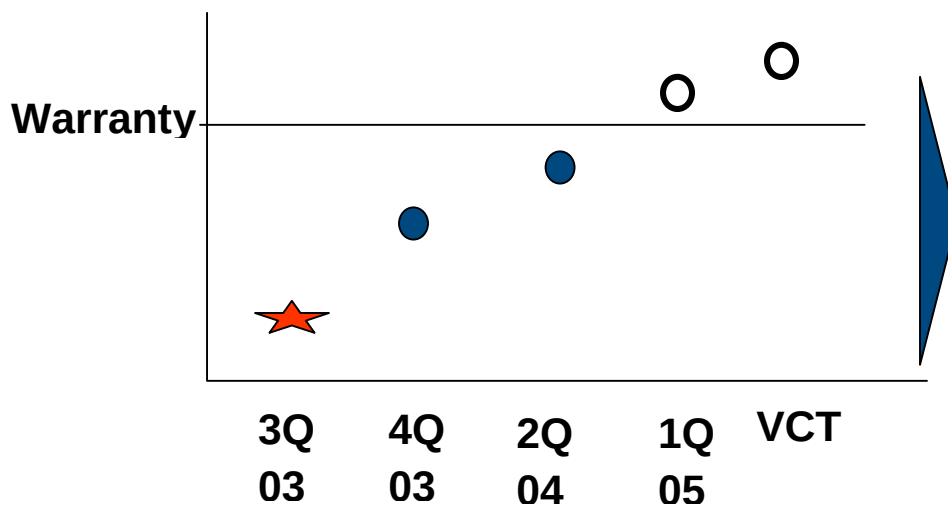
The Performix Pro 100 is the most powerful CT tube in the world. The power of the Pro 100 easily handles the most aggressive sub-second scanning needs and delivers more scanning capability than all other competitive tubes and systems combinations.

Reliability

Tube reliability significantly improved in 2004 and continues to improve. Over that period, the technology team has implemented several life-improving design changes to the casing, insert, cathode and anode. These changes have been released by model periods only after extensive internal customer-protocol based testing.

Testing

GE's accelerated testing procedures have proven to be good predictors of the actual CT Tube life that the installed base experiences. Our current accelerated testing indicates that the next model period design changes will result in a product whose life meets or exceeds the warranty period. One of the changes in this release, System Firmware Upgrade, is also expected to increase the life of tubes currently installed by up to 15%.



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Prerequisite FMIs

None

Related FMIs

FMI 25355 – HP80 and RT (Varian Tube) Quick Disconnect

FMI 25356 – HP80 (Varian Tube) Smart ID Rev2

FMI 25358 – HP100 & HP80 (Hercules Tube) kV Circuit Board Replacement

FMI 25360 – Helios Collimator Xray Scatter Filler Bar

Furnished Materials

Qty	Part #	Description
1	5135967	Software Kit (CDs)
1	5131410-100	FMI Doc.
1	2378261-100	LFC Doc.
1	46-302200P3	Rating Plate
1	2407441-100	LIGHTSPEED SERIES RELEASE
1	2369740-247	CT LIGHTSPEED SERIES CD
1	2376683	MCMaster CARR PN 6023A21 SOCKET 0.9375 IN DEEP IMPACT 0.500 IN DRV

Tool Required

Qty	Part #	Description	SW Version
1	5131410-100	FMI Doc.	N/A

Personnel Requirements

6 hours labor and 1 hours of travel

FMI Overview

All of the following procedures must be completed as part of this FMI:

- Load From Cold of the new software release
- Appendix A – Table Cradle Latch Inspection & Correction
- Appendix B - Table Cradle Drive Connections Inspection
- Appendix C - HP Table Grounding Inspection
- Appendix D - kV Circuit Board Inspection
- Appendix E - NGPDU Inspection

Procedure

Load From Cold Procedure

Use the LightSpeed GRE Software Installation Procedure document provided in this evaluation kit (2378261-100 Rev.15) and the steering guide below to install the new software on to your scanner.

Note: Make sure that system operation has been verified, all images have been archived, a Save System State has been generated, and a record of key system configuration parameter has been completed before you upgrade to the system software.

System Operation Verification

Put all the covers back and power up the system.

Set-up and perform at least one axial scan and one helical scan to verify that the system operated properly before beginning the software upgrade

Software Revision Check

Check the current version of CT Application Software on the system using the showprods command.

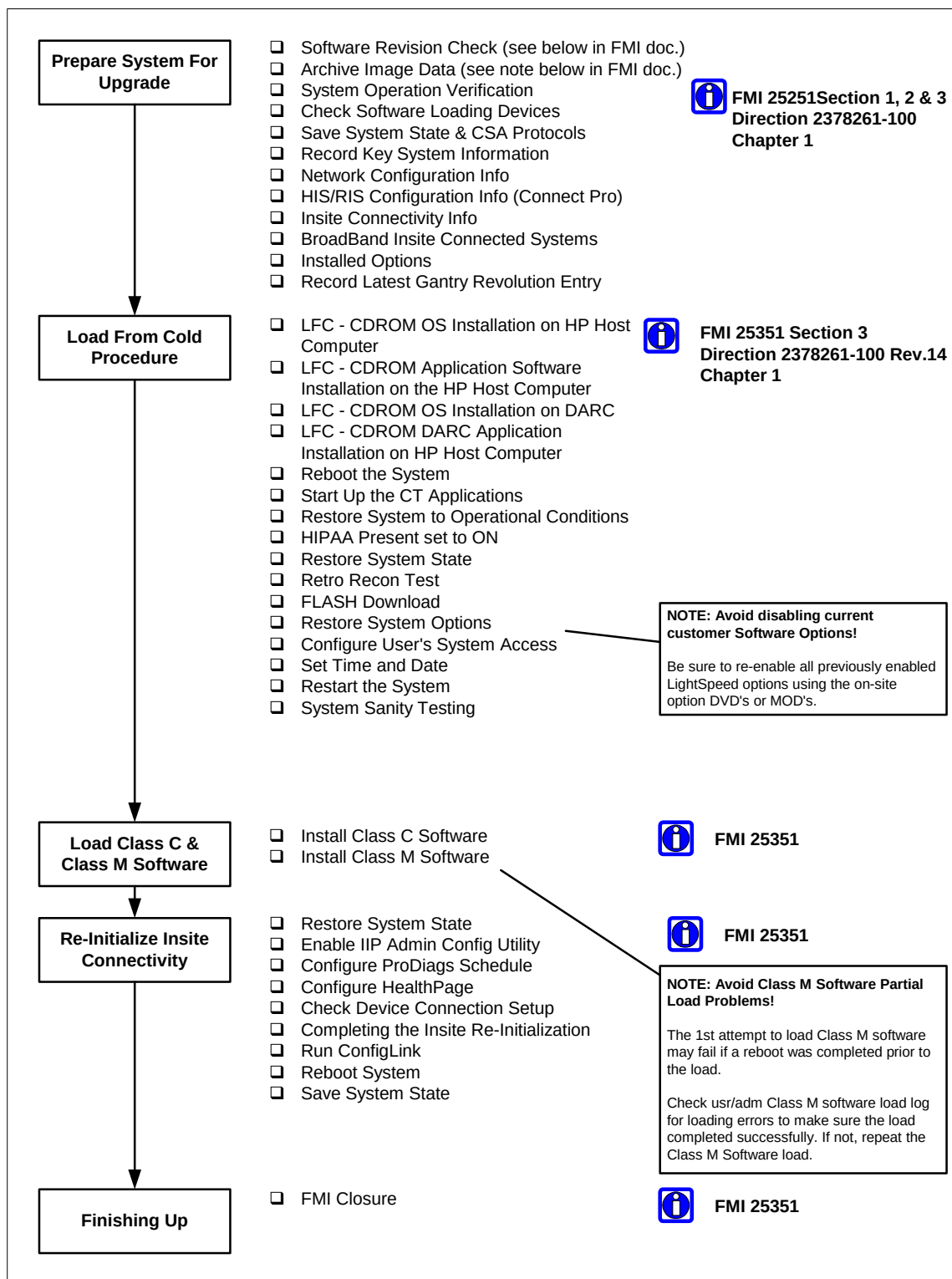
1. Open a shell.
2. At the prompt enter the following command

```
showprods | grep Helios_OC ENTER
```

Archive Image Data

The following procedures (Software Load From Cold) will destroy all image data on the system. Prior to starting the software load, make sure that the customer has reconstructed and archived all images before proceeding.

Steering Guide for GRE LFC



Load Class C and Class M Software

Notice: Avoid Partial Load of Class M Software

To ensure a proper load of the Class M Software, load Class C Software and immediately load Class M Software without rebooting the system between CD loads. The Class M Software CD load will fail to mount if not completed in this sequence. If the Class M Software install log exhibits Errors, usually a repeat attempt at loading Class M Software CD succeeds.

Load Class C Software

Load provided Class C software as follows:

1. Open a shell.
2. Enter root as superuser:
 - a. At the ctuser prompt, type: **su -** ENTER
 - b. At the Password prompt, type: **#bigguy** ENTER
3. Insert the Class C software CD into HP computer DVD Drive. Wait for the disk read to STOP.
4. Install the Class C software on the system. At the root prompt, type:
install.classC ENTER
5. Close Class C install reporting shell.
6. Verify Class C is properly loaded on the system.
 - a. At the root prompt, type: **cd /var/adm** ENTER
 - b. At the root prompt, type: **ls** ENTER
 - c. Find the install.classc.log file. If file found, this means the Class C software load completed.
 - d. At the root prompt, type:
more install.classc.log.<yyyymmddhhmmss>
Refer to Appendix A for an example of a successful Class C software load.
7. Remove the Class C software CD from HP computer DVD Drive.

Load Class M Software

Load provided Class M software as follows:

1. Open a shell.
2. Enter root as superuser:
 - a. At the ctuser prompt, type: **su -** ENTER
 - b. At the Password prompt, type: **#bigguy** ENTER
3. Insert the Class M software CD into the HP computer DVD drive. Wait for the disk read to STOP.

4. Install the Class M software on the system. At the root prompt, type:

install.classM **ENTER**

5. Verify Class M is properly loaded on the system.

a. At the root prompt, type: **cd /var/adm** **ENTER**

b. At the root prompt, type: **ls** **ENTER**

c. Find the install.classM.log file. If file found, this means the Class M software load completed.

d. At the root prompt, type:

more install.classM.log.<yyyymmddhhmmss>

Refer to Appendix A for an example of a successful Class M software load.

6. Remove the Class M software CD from HP computer DVD Drive.

Re-Initializing Insite Connectivity

This release of software contains a new IIP release (IIP 3.2-Linux). Changes in this IIP release are primarily in the following areas:

- Updates to support Multi-Tech modem firmware revision 8.19M
- Removal of IIP_DialPrefix ProDiags task
- Replacement of current iipHealthPage ProDiags task with a new Insite Data Mining task
- Auto Route to AutoSC in Level 5 setup

In addition, the Network Configuration Tab in the Insite Interactive Administration utility has been modified to eliminate the Use Default Gateway selection.

To ensure these IIP 3.2 features are correctly set, an Insite Checkout is NOT required. Because the basic Insite configuration information on the system should be correct from the original Insite Checkout, in most cases, it is only necessary to run ConfigLink via exiting the IIP Admin Configuration utility Insite Checkout Tab Screen.

Restore System State To Retrieve Class M Configuration Parameters

Once Class M software has been loaded, you must repeat a Restore System State to recover all previous Insite and Class M configuration parameters.

Reference: Direction 2378261-100, GRE Software Installation

Section 3.8 Restore System State

NOTE: Make sure that the Key System Information recorded in Section 1 (Insite Connectivity Info) is correctly set as recorded. If not, use the IIP Admin Configuration utility to reset Network Configuration Info as described in the following procedures.

Enable Insite Interactive Platform Configuration Utility

See Figure 2, Insite Interactive Platform Configuration Utility Gaining Access and Selecting ProDiags Utility.

Enable the Insite Interactive Platform Configuration utility as follows:

1. Open a shell.
2. Enter root as superuser:
 - a. At the ctuser prompt, type: `su -` **ENTER**
 - b. At the Password prompt, type: `#bigguy` **ENTER**
3. Launch the IIP Admin Configuration utility. At the root prompt, type:
`iipadmin config` **ENTER**
The system displays the Insite Interactive Platform Configuration utility with the Proprietary Agreement Acceptance Screen.
4. Click on **ACCEPT** three times to display the Information Screen.
5. Click on the **ProDiags** Tab to display the ProDiags Screen.
6. Click on the **Custom** button to launch the independent ProDiags utility as shown in Figure 3.

Figure 2
Insite Interactive Platform Configuration Utility

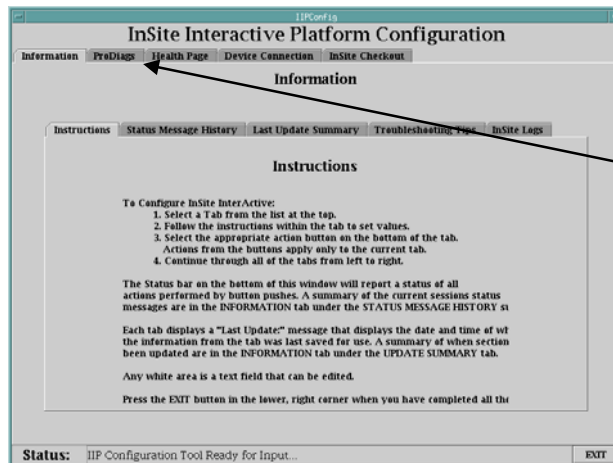
Gaining Access and Selecting ProDiags Utility



Proprietary Agreement Screen

1.

Click **Accept** Three Times to gain access to the Insite Interactive Platform Configuration Utility



Information Screen

2.

Click on the **ProDiags** Tab to display the IIP Admin Config Utility ProDiags Screen



ProDiags Screen

3.

Click on the **Custom** button to launch the independent ProDiags Utility

Configure ProDiags Schedule

With this release of software, the IIP 3.2 package contains a new data collection package that pushes gesyslog data and a basic system configuration file to the On Line Center RDA

Database. This IDM package contains two new ProDiags tasks (idm and idm_send_data) that must be configured to match the site's operational usage.

The default ProDiags schedule provided in this release should be adequate for most systems. However, we recommend that you review this schedule and update according to your particular site's normal operation.

Updating ProDiags Scheduled Tasks

See Figure 3 and refer to detailed steps on the following page.

Figure 3

ProDiags Graphical Utility

Schedule Tab – Modifying Scheduled Tasks

1. Click on **Hour** and **Minute** fields of idm_send_data task and select an hour and minute from the pull down lists between 9 AM and 5 PM. After each selection, click **Yes** on the "Data has been modified. Please confirm it" pop up.

2. Click on **Hour** and **Minute** fields of idm task and select an hour and minute from the pull down lists to schedule 5 minutes before the idm_send_data task. After each selection, click **Yes** on the "Data has been modified. Please confirm it" pop up.

3. When idm and idm_send_data Hour and Minute selections are complete, click on **Apply** to finalize the schedule updates. Click **Yes** on the "Are you sure you want to apply changes?" pop up.

4. Close the ProDiags utility browser by clicking on **Close** from the upper left browser window pull down menu.

Note: If

Use the following sequence to update the scheduled time (Hour and Minute) of the idm and idm_send_data tasks:

ProDiags idm_send_data task

- Click on the **idm_send_data** task **Hour** field.

- From the pull down list, select an hour that the system is normally on (typically between 09:00 and 17:00).
- On the "Data has been modified. Please confirm it" pop-up, click on **YES**.
- Click on the **idm_send_data** task **Minute** field.
- From the pull down list, select a minute (preferably an odd minute such as 3, 14, 22, etc).
- On the "Data has been modified. Please confirm it" pop-up, click on **YES**.

ProDiags idm task: Set up the idm task to run five minutes before the idm_send_data task.

- Click on the **idm** task **Hour** field.
- From the pull down list, select an hour that sets the idm task to run five minutes before the idm_send_data task.
- On the "Data has been modified. Please confirm it" pop-up, click on **YES**.
- Click on the **idm_send_data** task **Hour** field.
- From the pull down list, select a minute that sets the idm task to run five minutes before the idm_send_data task.
- On the "Data has been modified. Please confirm it" pop-up, click on **YES**.

When completed with the idm and idm_send_data modifications, click on the **APPLY** button at the bottom of the the Schedule Tab. On the "Apply change confirmation" pop-up, click on **YES**.

NOTICE: If the schedule change failed due to a permissions denied error, execute the following change-mode command to make the file named bg_schedule_data read/writable:

- With the ProDiags GUI still open, open another shell
- Become root user:
At the ctuser prompt, type: **su -** **ENTER**
At the Password prompt, type: **#bigguy** **ENTER**
- Set Permissions for bg_schedule_data:
type: **cd /usr/g/insite/ProDiags.schedule**
type: **chmod 777 bg_schedule_data**

Repeat the Apply click on the ProDiags Schedule Tab. The update should occur without failures.

Close the ProDiags utility browser by selecting **Close** from the upper left browser window pull down menu. The ProDiags browser should close and the Insite Interactive Platform Configuration utility should display the HealthPage Tab as shown in Figure 4.

Configure HealthPage

After completing the ProDiags setup, the IIP Admin Configuration utility automatically switches to the HealthPage Tab as shown in Figure 4.

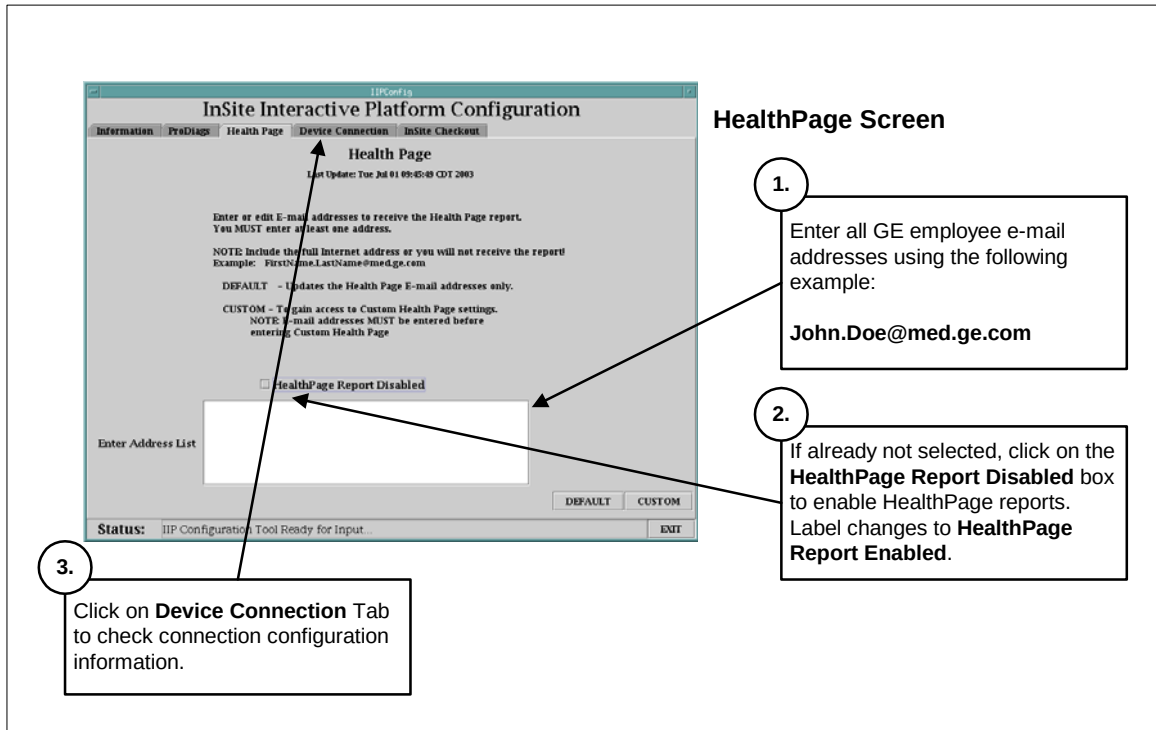
The Restore State process (after loading Class M software) restored the following information for the HealthPage Reports:

- All e-mail addresses of targeted recipients that were previously loaded
- HealthPage report contents that was previously configured

- HealthPage report schedule that was previously configured

Unless you have recent changes to any of these three items, you can skip the HealthPage Tab of the IIP Admin Configuration utility and select the Device Connection Tab. If you have changes that you want to make (e-mail addresses, report content, report schedule), follow the instructions on the IIP Admin Configuration utility.

Figure 4
Insite Interactive Platform Configuration Utility
HealthPage Tab



Check Device Connection Setup

Modem Setup

When selecting the Device Connection Tab, the IIP Admin Configuration utility automatically displays the Modem Setup Screen as shown in Figure 5.

If your system was configured for Modem connectivity, the Restore State process (after loading Class M software) restored all the Modem setup parameters. To verify, refer to Section 1 of this FMI where you recorded Insite Connectivity Info.

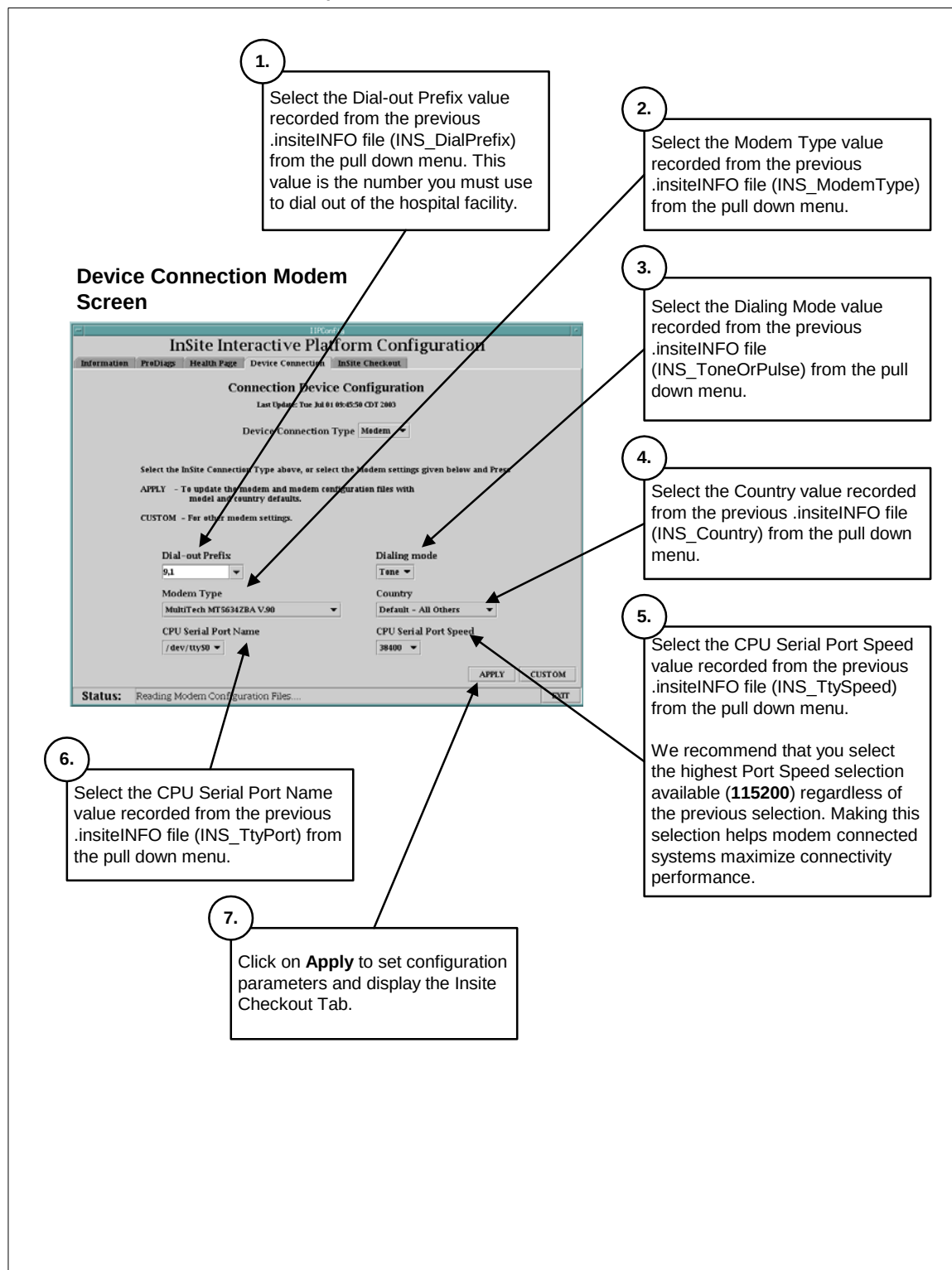
- If there were no Modem parameters in the .insiteINFO file but there were BroadBand parameters, skip to the Check BroadBand Device Connection Setup procedure.
- If there were Modem parameters in the .insiteINFO file but the parameters displayed in the Modem Setup Screen do not match what you previously recorded, enter the correct values as required.

NOTE: The Modem configuration parameters should be pre-loaded from the Restore State process. Do not change settings unless these values were not properly restored.

If required, follow the instructions in Figure 5. The default values are usually the correct values or most commonly used. Parameter selections that do not require customization are:

CPU Serial Port Name

Figure 5
Insite Interactive Platform Configuration Utility
Device Connection Tab – Modem Setup



BroadBand Setup

If your system was previously BroadBand Insite connected, click on the Network button of the IIP Admin Configuration utility Device Connection Tab to display the Network Setup Screen as shown in Figure 6.

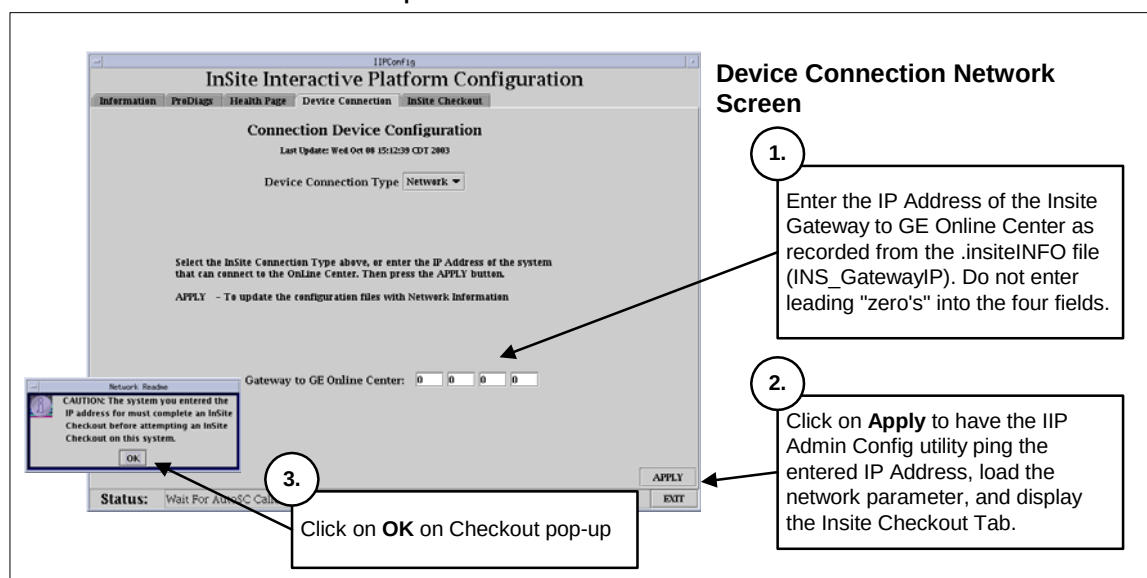
If your system was configured for BroadBand connectivity, the Restore State process (after loading Class M software) restored all the BroadBand setup parameters. To verify, refer to Section 1 of this FMI where you recorded Insite Connectivity Info.

NOTE: The BroadBand Network configuration parameters should be pre-loaded from the Restore State process. However, the on-site FE who performed the original Insite Checkout of the system may have used the hospital's imaging LAN default gateway IP Address (selection was available in the previous release of IIP).

DO NOT USE THE HOSPITAL WORKING LAN DEFAULT GATEWAY IP ADDRESS AS THE INSITE GATEWAY IP ADDRESS OTHER THAN AS A TEMPORARY MEANS TO CONNECT TO THE ON LINE CENTER!!

To find the correct Insite Gateway IP Address, use the contacts listed in Section 1.10, BroadBand Insite Connected Systems.

Figure 6
Insite Interactive Platform Configuration Utility
Device Connection Tab – Network Setup



Completing the Insite Re-Initialization

Follow the instructions provided in Figure 7 to exit the IIP Admin Configuration Utility.

Figure 7

Insite Interactive Platform Configuration Utility

Insite Checkout Tab – Exiting the IIP Admin Configuration Utility

Insite Checkout Screen

1. Click on the Exit button of the IIP Admin Config utility to exit the utility.

2. Click on **OK** on the Exit Confirmation pop-up to exit the IIP Admin Config utility and run the ConfigLink utility.

3. After the shell displays the root prompt, reboot the system to load Insite connectivity parameters. As the system is coming up, you should see the iLinq icon (if the original Checkout included iLinq) as a selectable desktop on the right console monitor.

Run ConfigLink

Run the Insite **ConfigLink** script to populate all configuration information required to fully restore connectivity parameters on the system.

1. Open a shell.
2. Enter root as superuser:
 - a. At the ctuser prompt, type: **su -**
 - b. At the Password prompt, type: **#bigguy**
3. Change directory to Insite. At the root prompt, type:
cd /usr/g/insite
4. Run the ConfigLink script. At the root prompt, type:
ConfigLink

Wait for the shell to return the root prompt to ensure the **ConfigLink** process completes. When complete, exit root and close the shell.

Reboot System

To ensure all Insite configuration parameters are properly loaded into the operating system (route tables, etc.), perform a system reboot.

1. Click on the pink Shutdown button.
2. From the Shutdown selection pop-up, click on , then , to reboot the system.

Save System State

At the completion of the Load From Cold Process and Insite Re-Initialization process, ensure all system state parameters are saved using the Save System State Utility.

Reference: Direction 2378261-100, GRE Software Installation

Section 3.2 Save System State

Finishing Up

Note: Before performing this step, verify that you completed all checks in Appendix A through E.

Perform the Gantry Rotation Safety check, found in the System Service Manual. If you cannot find the procedure, click on the link from the Tube Change procedure, as the manual may have a bug where it may not be displayed.

Also perform an application scan to ensure proper system operation before turning the system over to the customer.

Appendix A - Table Cradle Latch Inspection & Correction

PROBLEM HISTORY:

There have been many complaints about HP Tables in the Field. The problem occurs when the Table is at the home location (Fully Retracted) the Home Latch Assembly Captures the Cradle at the Home Location. If a patient is loaded onto the Cradle and if the patient is pulled / slid towards the Gantry, this results in pressure being applied to the Home Latch Mechanics. As a result of this pressure or pre-load on the Home Latch Hook, frequently the Home Latch will fail to Mechanically Respond when its Solenoid is energized. If the Home Latch becomes Mechanically Stuck in this condition attempts to drive the Cradle into the Gantry result in Cradle Drive Roller Assembly Slippage.

You will hear the Cradle Drive Motor spinning rapidly but the Cradle does not move.

Many of the Operators are aware of this Condition and have learned to pull the Cradle Back to the home position after loading the patient on the table. As an enhancement to the table operation the Firmware has been changed as follows. If the Cradle location is sensed as being with-in a few millimeters of the home location and the Gantry push buttons are pressed to request Cradle Extend / In motion then Firmware will take the following actions. First a command is sent to the Table to move it to the Home Location, and then after this move is complete (only a few mm) a second command is sent to the table to generate the Extend into Gantry motion.

The Firmware is commending the cradle to first backup prior to extending the Cradle when the Cradle is near the home Latch Position.

As a Note: Since there are two moves being conducted in rapid succession you may hear the Home Latch activate and release twice (as designed).

For this M4 Software / Firmware modification to prove effective in resolving this issue it is important to have the Home Latch properly adjusted in relation to the Characterized Home Location. Hence the procedure(s) as listed below.

When:

This inspection must be completed after the M4 software load.

Inspection type:

Functional:

Follow the functional test procedure.

Tools and material required

- Standard FE tool kit

Time: One engineer ½ hour

System Level Hazards

Electrical Hazard 

Mechanical Hazard 

Tag and Lock Out 

Follow all safety procedures and use all required personal protection equipment that are applicable for this procedure. These procedures can be found in the system service manual.

Follow the appropriate LOTO procedure for this procedure:

Functional Test Instructions

After loading M4 software

- Drive the cradle to the home position using the gantry control panel.
- Landmark the cradle (internal landmark) when at the home position.
- Manually push the cradle forward until it is forced into contact with the cradle latch (can not push it any further forward).
- With the cradle pushed up against the latch. Read the gantry display.
- If the gantry display (cradle) reads 1.0 to 2.0 then it is likely that the latch is mechanically within tolerance (if greater than 2.0, then the latch will need to be adjusted).
- With the cradle still pushed up against the latch, go over to the gantry control panel and drive the cradle forward (extend) do not use the Fast Button. If the cradle backs up and then drives forward, the latch is within tolerance. If the cradle only attempts to drive forward, then the latch is out of tolerance and must be adjusted.

Repair Instructions

Adjusting Latch:

- The latch must be adjusted so that it catches when the cradle is 0.75 to 2.0 mm from home.
- The gantry display only reads in increments of 0.5mm so it is not the best method for determining where the latch is catching.
- Open up a Unix Shell and type the following:

```
rlogin tgp
```

```
uifShow
```

Look for "cradle absolute". It is in micrometers (divide by 1000 to get mm). At home this will be 0. You must re-type to refresh / get an updated reading.

Appendix B- Cradle Drive Connections Inspection

The following CT Equipment is affected:

Product	Model #
All HP Tables	2335179

Table 1 – Table Model

Purpose

The purpose of this inspection is for all HP Tables model 2335179, cradle drive cables for damage due to poor cable management on some tables. This Inspection will have the engineers inspect cables for damage, make cable repairs as needed and secure cable harness to the table frame.

Tool Required

Qty	Description
1	This FMI doc.
1	Standard FE Tool Bag
1	Electrical Tape
1	Flash Light
1	

Cradle Inspection and Repair Procedure

Before You Begin

Follow all required safety procedures for this product when performing this procedure. LOTO as required. CT Service training is required to perform this procedure. Additional service procedures are available in the CT system service manual.

Warning: Make sure before starting this Inspection that all power to the scanner is remove using lockout / tagout.

Inspection:

Table Cover Removal

- 1) Remove the table side covers using the two mounting screws. see figure 1
- 2) Remove the cradle drive bottom service panel using the two thumb screws. see figure 2
- 3) Remove the cradle front cover bezel using the two hex head screws on each side of the cradle. see figure 3



Figure 1 side cover



Figure 2 drive cover



Figure 3 roller cover

Cradle Cables Inspection

- 1) On the cradle drive assembly check that all wires are secured to the frame. Wires should not come in contact with moving parts. See figure 4 & 5 Remove roller debris as needed.
- 2) On the Motor and encoder assembly all control cables should be neatly secured to the cradle frame. See figure 6. All electrical connection should be secure and tight.



Figure 5 control cable rubbing on belt



Figure 4 cradle drive assembly



Figure 6 encoder and cradle drive motor cables

Secure Cables as Needed

- 1) Locate the gray cable shown in figure 5 with the four wires.
- 2) Secure this cable as shown in figure 7 to the existing cable clamp. If your clamp is too small use the larger clamp in the kit.
- 3) Use the supplied cable ties to dress all other cables and wires. Your table cables should look like the cables in figure 8 when finished. The two black motor cables are stiff and difficult to position, be careful not to break these two cables.

Caution:

These four wires can get caught in this belt assembly and break!

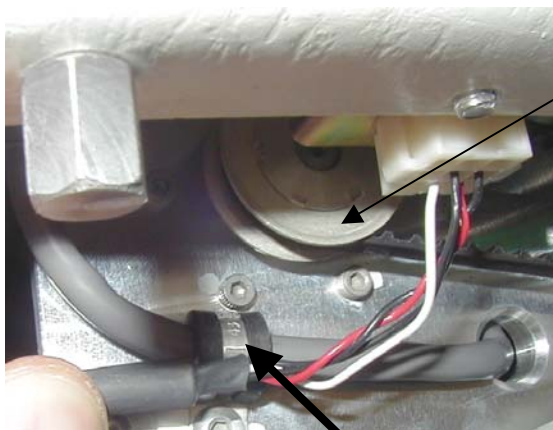


Figure 7 Clamp this cable here



Figure 8 Secure cables as showed

OPTIONAL REPAIR PROCEDURE

For Broken or Damaged Cradle Clutch Cable

Tools required

- ✓ wire stripper
- ✓ pin removal tool
- ✓ 6" shrink tubing
- ✓ Amp male pins
- ✓ crimping tool
- ✓ Drawing 2347447DDW
- ✓ Repair Kit
- ✓ Heat gun

Repair Procedure for Encoder Clutch Cable

- 1) It is possible to have a damaged cable as shown in figure 5. If your cable is damaged you can re-terminate this cable using the pins supplied in the FMI kit. *
- 2) Note: This plug should be unplugged from the top first. Then record the pin locations in the plug. Use a pin removal tool to remove all four pins. Refer to the plug pin out drawing 2347447DDW. **Note:** this is not a one to one pin out.
- 3) Remove any existing cable clamps to gain enough free cable to reach the connector.
- 4) Strip the cable back enough to expose 3" of the red and white wires. Follow the "all other" procedure found on print 2347447. Use the shrink tubing to secure the braid shield to the cable.
- 5) Strip each of the four wires enough to fit the supplied pins and crimp all four pins. The pin connections should be secure to the touch making good electrical contact with each wire. Using a DVM OHM out the wires between cradle end A33 pins 1;4;2;5 to table base end A2 pins 1;2;3;4.
- 6) Using the information from step 2, or the print 2347447, Insert each pin into the A 33 connector. Make sure that all pins are securely seated in the connector, then reattach the connector.
- 7) Replace all removed cable clamps, and check that no wires or cables are rubbing on any moving components.

* If your cables are damaged beyond repair, call the OLC for support in ordering and replacing the table harness.

Repair Procedure for Cradle Motor Drive Cables –

Danger! High Voltage Present

Replace the **Cradle Drive** assembly

Repair Procedure for Cradle Encoder Cable

Replace the **Encoder** assembly

Appendix C - HP Table Grounding Inspection

Purpose

The purpose of this procedure is to inspect the grounding connection for all HP Tables model 2335179. During installation some of the ground cables were connected to the wrong ground point.

Tool Required

Qty	Description
1	This FMI doc.
1	Standard FE Tool Bag

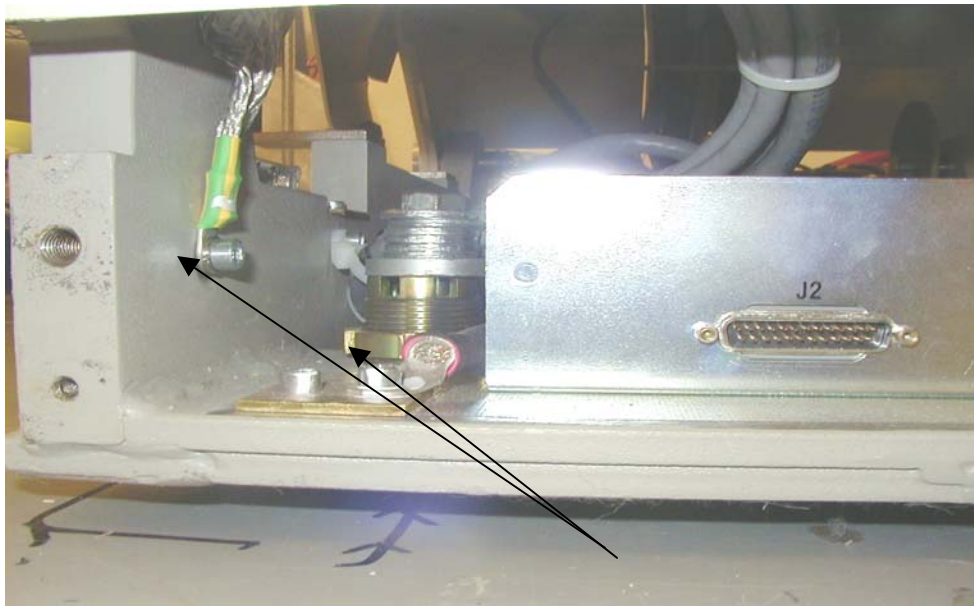
Before You Begin

Follow all required safety procedures for this product when performing this procedure. LOTO as required. CT Service training is required to perform this procedure. Additional service procedures are available in the CT system service manual.

Warning: Make sure before starting this Inspection that all power to the scanner is remove using lockout / tagout.

Inspection:

1. With covers already removed please inspect the ground connection for the table and correct it if necessary. See the correction below.



If your table has the main ground connected to the table frame as shown, remove this ground, and attach this to the main table ground buss on the footswitch assembly.

Appendix C - Completion

Cover Replacement

1. Replace all removed covers. When doing so be sure that no cables are binding or pinched.
2. Remove LOTO devices and restore system power

Functional Test

1. Complete a system functional test. The table should function properly.

If the table is not functioning properly, remove the table covers and recheck all cable connections.

Appendix D - PPC kV control board inspection

Purpose

The spit detection circuit in the PPC board 2351200-11 rev 001 is too sensitive at high kV. At certain techniques, the natural ripple in the kV could cause false and repetitive spit detection leading to scan aborts. The scan aborts due to false spit detection can lead to unnecessary x-ray exposure to the patient and false indication of spitting x-ray tube.

The defect is only in PPC board 2351200-11 rev 001. PPC boards prior to 2351200-11 rev 001 do not present this problem and later board designs including 2407577, do not exhibit the failure.

VERY IMPORTANT: *Please inspect the rev. of the board your scanner is using and, if wrong rev of the board is used please verify that you are on the affectivity list for FMI 25358. If you scanner is not on the affectivity list please call or email Alex Sender at 414-807-3187, Arkadiusz.Sender@med.ge.com to be added to the affectivity for this FMI.*

Tool Required

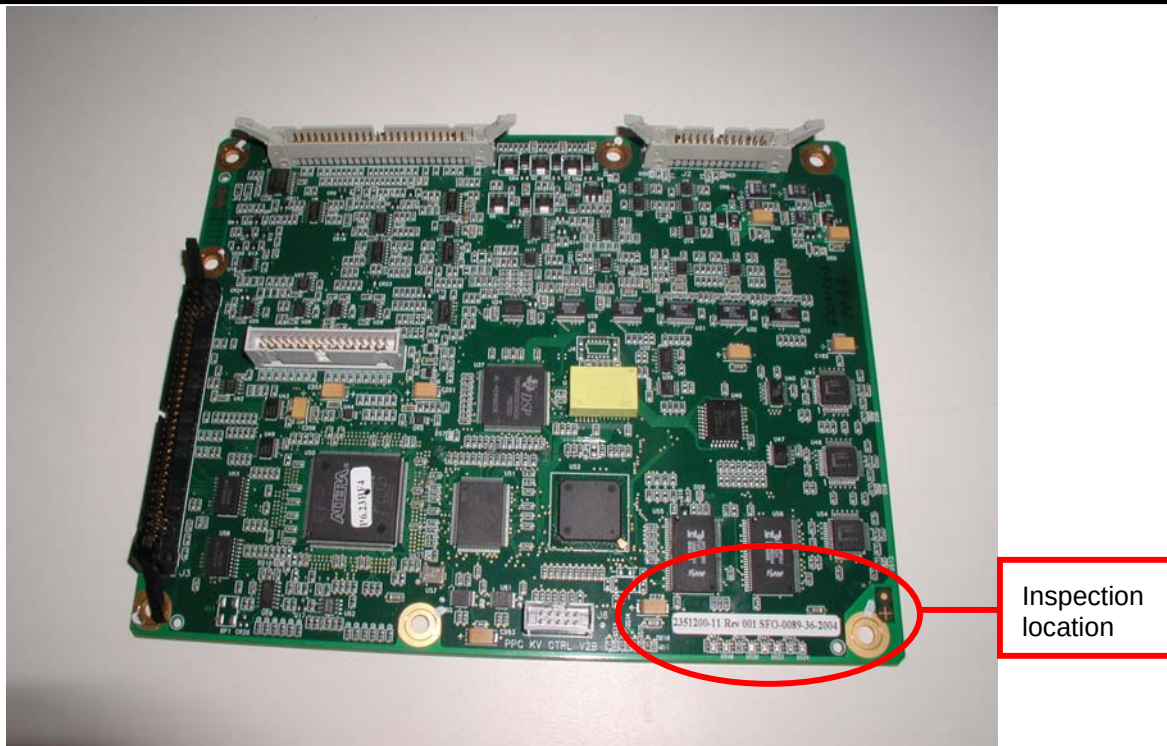
Qty	Description
1	Standard FE Tool Bag

Before You Begin

Follow all required safety procedures for this product when performing this procedure. LOTO as required. CT Service training is required to perform this procedure. Additional service procedures are available in the CT system service manual.

Inspection:

1. Remove gantry side cover and power off 120Vac
2. Remove cover on JEDI inverter to access PPC Circuit Board
3. Inspect label on PPC board



4. If the board part number is 2351200-11 rev 001 it needs to be replaced. If this is a case please refer to the purpose statement for this inspection of further action.

5. Close the Jedi cover and put back the scanner covers and apply power to the system. You can operate the scanner as normal until you get the FMI25358.

Appendix E – NGPDU Inspection

Purpose

CAUTION **ELECTRICAL HAZARD!** Please be assured that you have understood all the safety guidelines that you have to follow during your work. Please Turn OFF the Main Power Distribution Supply to CT system.

The spit detection circuit in the PPC board 2351200-11 rev 001 is too sensitive at high kV. At certain techniques, the natural ripple in the kV could cause false and repetitive spit detection leading to scan aborts due to too many spits. This is a risk for the patient. The scan aborts due to false spits could also lead to replament of a tube that is not defective This defect in PPC board 2351200-11 is a regression. PPC boards prior to 2351200-11 do not present this problem and the problem has been corrected in later board designs including 2407577

Tool Required

Qty	Description
1	Standard FE Tool Bag

Before You Begin

Follow all required safety procedures for this product when performing this procedure. LOTO as required. CT Service training is required to perform this procedure. Additional service procedures are available in the CT system service manual.

PROCEDURE

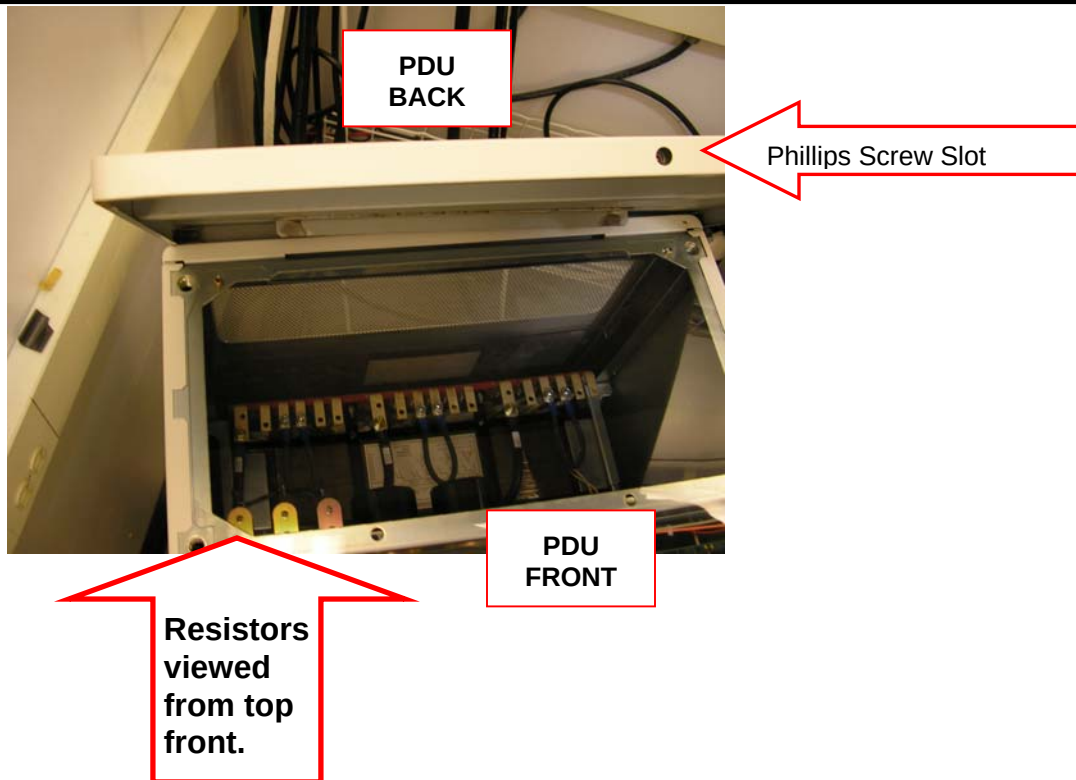
1 PREPARATION

Understand and follow the Safety Guidelines Manual.

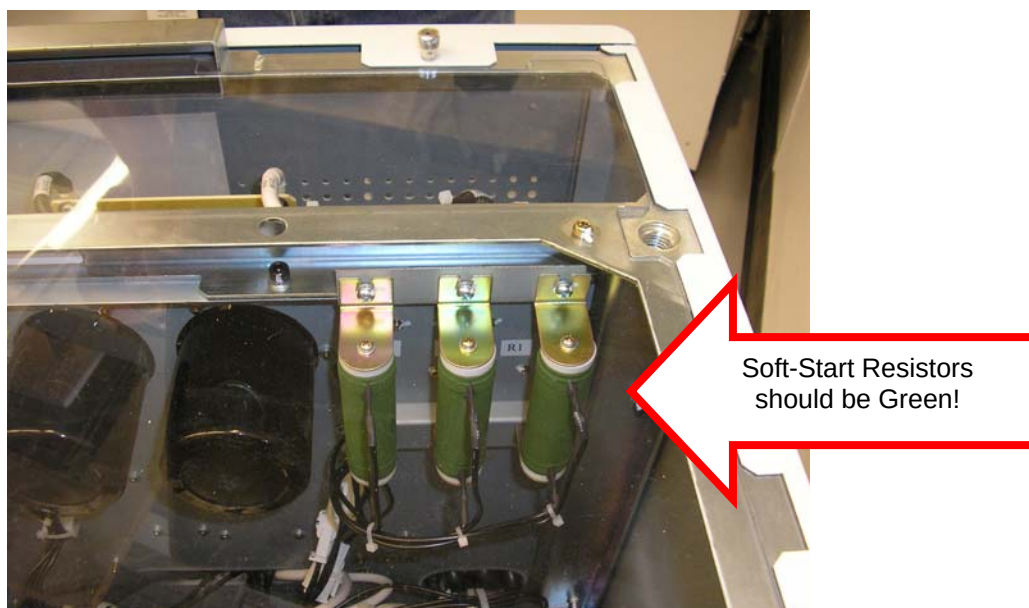
- 1.) Switch OFF the OC main power.
- 2.) Switch OFF the breaker in the Power Distribution Box (hospital supply). Note: If there is UPS, Switch OFF the UPS.

2. VISUALLY INSPECT NGPDU SOFT-START RESISTORS!

1. Remove cover to NGPDU, by using Phillips head screw driver to loosen screw in front cover.
2. Lift cover off cabinet and place safely off to the side.



3. Examine the three (3) soft-start resistors, if resistors are green in color, resistors are the correct size. If they are white or silver, then please order FMI 22901, part #2405313 and return to install FMI when kit arrives.



4. Replace NGPDU cover and return system to normal operation.